AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 10 · STUDENT LESSON

Evaluation pipelines

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

Before you start

Recall Integration tests and fixtures. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of evaluation pipelines. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of evaluation pipelines.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Integration tests and fixtures. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Testing pure components. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Configuration and secrets. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of evaluation pipelines

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Integration tests and fixtures. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

This week's end goal

Explain evaluation pipelines using the deciding step in this fictional service.

Notice the evidence

> Investigate the supplied Evaluation pipelines case. Separate observations, given facts and assumptions before choosing an explanation.

Evaluation pipelines preserve case versions, procedure versions and **raw outcome**s before producing metrics. A rerun must identify what changed.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Evaluation pipelines. Connect each labelled input, deciding step and result.

Load versioned cases, validate expected outcomes, execute the frozen system, store case-level results and compute declared denominators. Reject missing results rather than silently counting them as success.

Evaluation pipelines — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Six cases are planned; five run, four pass and one fails. One is skipped.	Load versioned cases, validate expected outcomes, execute the frozen system, store case-level results and compute declared denominators. Reject missing results rather than silently counting them as success.	Completed-case rate 4/5=80%; coverage 5/6. Do not report 4/6 as completed accuracy without declaring that separate policy; skipped remains unassessed.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

Compare the cases

> Compare the supplied cases for Evaluation pipelines. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Six cases are planned; five run, four pass and one fails. One is skipped. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Change one thing in the pictured case

Change one thing

Change: Remove the shared failed third case after scoring.

Predicted result

Prediction: Rates become 3/3 and 2/3, but this selectively edited evaluation is misleading.

Why the result follows

Reason: The population changed after seeing failures, concealing a weakness.

Words that unlock the idea

> Use the vocabulary of Evaluation pipelines to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Pipeline manifest	Versions and expected stages of an evaluation run
Coverage denominator	The planned cases eligible for evaluation
Raw outcome	The stored case-level result before aggregation

> **Use two terms to explain a step in this week's example.**

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Evaluation pipelines, predict a step, then compare with its explanation.

Six cases are planned; five run, four pass and one fails. One is skipped.

Completed-case rate 4/5=**80%**; coverage 5/6. Do not report 4/6 as completed accuracy without declaring that separate policy; skipped remains unassessed.

Worked explanation · Evaluation pipelines

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Evaluation pipelines

Four cases through a scoring pipeline. Two fixed versions answer the same four questions with frozen correct-or-incorrect labels.

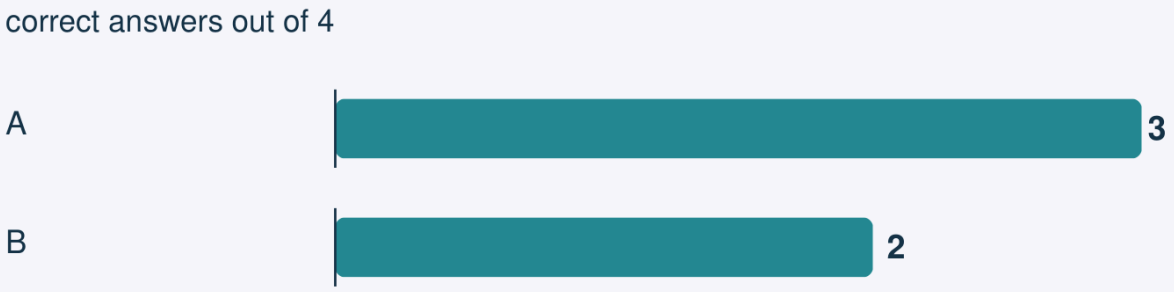
The facts we will use

Supplied facts

A scores [1,1,0,1]; B scores [1,0,0,1]. Each 1 means agreement with the reference answer.

AI ACADEMY / AI-6/10

Correct answers on four shared cases



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-6/10

Evaluation pipelines

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Freeze cases

Keep questions and reference labels unchanged for both versions.



2. Compute rates

A is 3/4=75%; B is 2/4=50%.



3. Inspect failures

Case two separates the versions; case three fails for both.

CHANGE ONE THING

Remove the shared failed third case after scoring.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Freeze cases	Keep questions and reference labels unchanged for both versions.
2. Compute rates	A is 3/4=75%; B is 2/4=50%.
3. Inspect failures	Case two separates the versions; case three fails for both.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A pipeline preserves comparable cases, version labels and scoring. Report the denominator and disagreements alongside percentages.

What this example does not establish

Keep the stated limit

Four fictional cases illustrate a procedure, not broad quality or a stable future success rate.

Keep the stated limit

Read the labelled picture: Calculate A's and B's correct rates on all four cases. Identify their shared failure and explain why removing that case after scoring changes the honesty of the comparison.

Keep the stated limit

Change one thing: Remove the shared failed third case after scoring. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Evaluation pipelines claim and explain a repair using the supplied evidence.

Missing evaluations can bias results; neither completion rate nor pass rate should hide the other.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: Missing evaluations can bias results; neither completion rate nor pass rate should hide the other.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

Practise together

> Practise Evaluation pipelines with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Evaluation pipelines”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Evaluation pipelines. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Calculate A’s and B’s correct rates on all four cases. Identify their shared failure and explain why removing that case after scoring changes the honesty of the comparison.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Evaluation pipelines. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Six cases are planned; five run, four pass and one fails. One is skipped. Show the intermediate steps and give the conclusion with its scope.

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Evaluation pipelines. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: Missing evaluations can bias results; neither completion rate nor pass rate should hide the other.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Grounded Learning Service. Keep an annotated decision and the evidence supporting it.

Add a labelled evaluation pipelines **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of evaluation pipelines.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **My limited claim and a useful next check**

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Evaluation pipelines case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Evaluation pipelines — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Six cases are planned; five run, four pass and one fails. One is skipped.	Load versioned cases, validate expected outcomes, execute the frozen system, store case-level results and compute declared denominators. Reject missing results rather than silently counting them as success.	Completed-case rate 4/5=80%; coverage 5/6. Do not report 4/6 as completed accuracy without declaring that separate policy; skipped remains unassessed.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Eight planned cases, six complete, five pass, one fail. Give observed pass rate and coverage.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Eight planned cases, six complete, five pass, one fail. Give observed pass rate and coverage.

> **My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Evaluation pipelines, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Four cases through a scoring pipeline. Two fixed versions answer the same four questions with frozen correct-or-incorrect labels.

By the end, I can	How I will show it
Explain the mechanism and terminology of evaluation pipelines	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Evaluation pipelines without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of evaluation pipelines

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Four cases through a scoring pipeline. Two fixed versions answer the same four questions with frozen correct-or-incorrect labels.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Case two separates the versions; case three fails for both.

1. Freeze cases

2. Inspect failures

3. Compute rates

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Inspect failures Case two separates the versions; case three fails for both. A pipeline preserves comparable cases, version labels and scoring.

Report the denominator and disagreements alongside percentages.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Remove the shared failed third case after scoring. Which conclusion is supported by the supplied facts?

1. Rates become 3/3 and 2/3, but this selectively edited evaluation is misleading.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Rates become 3/3 and 2/3, but this selectively edited evaluation is misleading. The population changed after seeing failures, concealing a weakness.

Repair a missing building block

Review Integration tests and fixtures (ai-6/9). Goal: Explain the **mechanism** and terminology of integration tests and fixtures. Short practice: Follow request N through the fixture and adapter.

Which source and item should appear in the reply, and what precisely fails when the adapter returns M?. Return to this week and explain what you changed.

Review Testing pure components (ai-6/7). Goal: Explain the **mechanism** and terminology of testing pure components. Short practice: Predict the formatter **output**s for 7 and 12, including spacing.

Why does **input** 0 need rejection under this contract rather than an ordinary formatted response?. Return to this week and explain what you changed.

Review Configuration and secrets (ai-6/4). Goal: Explain the **mechanism** and terminology of configuration and secrets. Short practice: Point to the public setting and the protected credential.

What changes when maximum_words becomes 60, and which information must never appear in the startup log?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

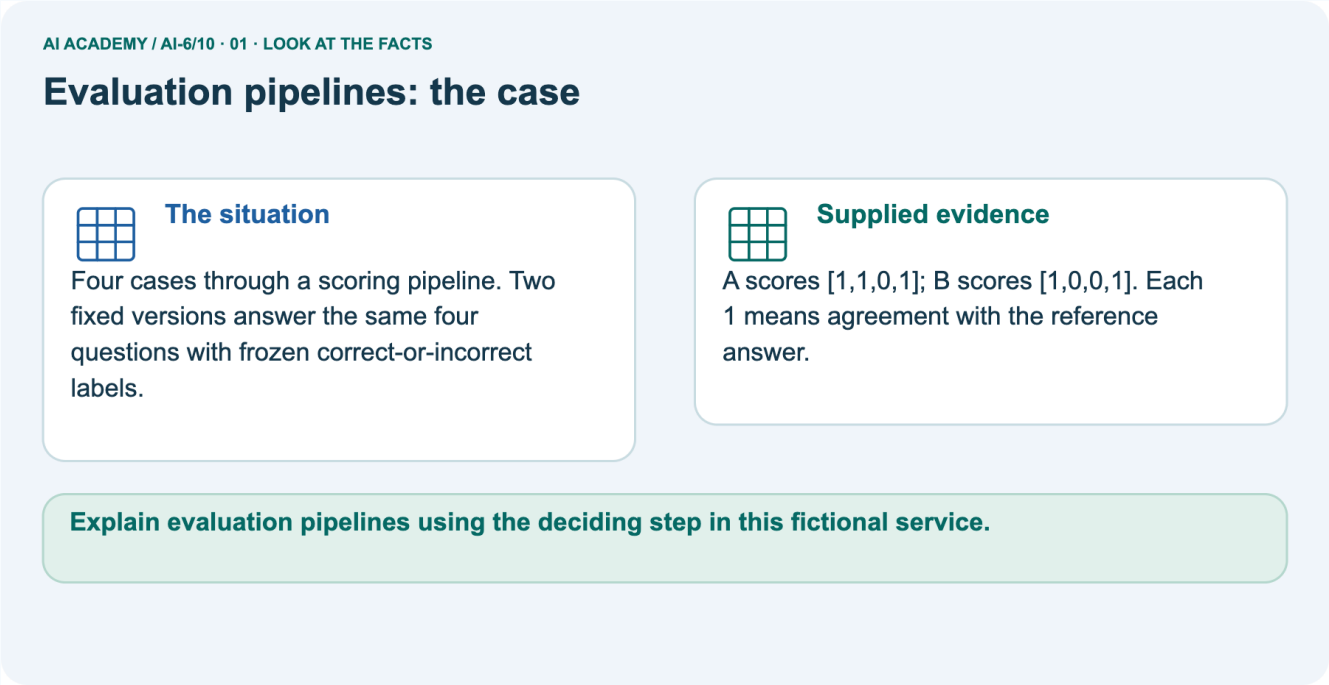
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Evaluation pipelines graphic using its labels, deciding evidence and relevant limitation.

01. Evaluation pipelines: the case



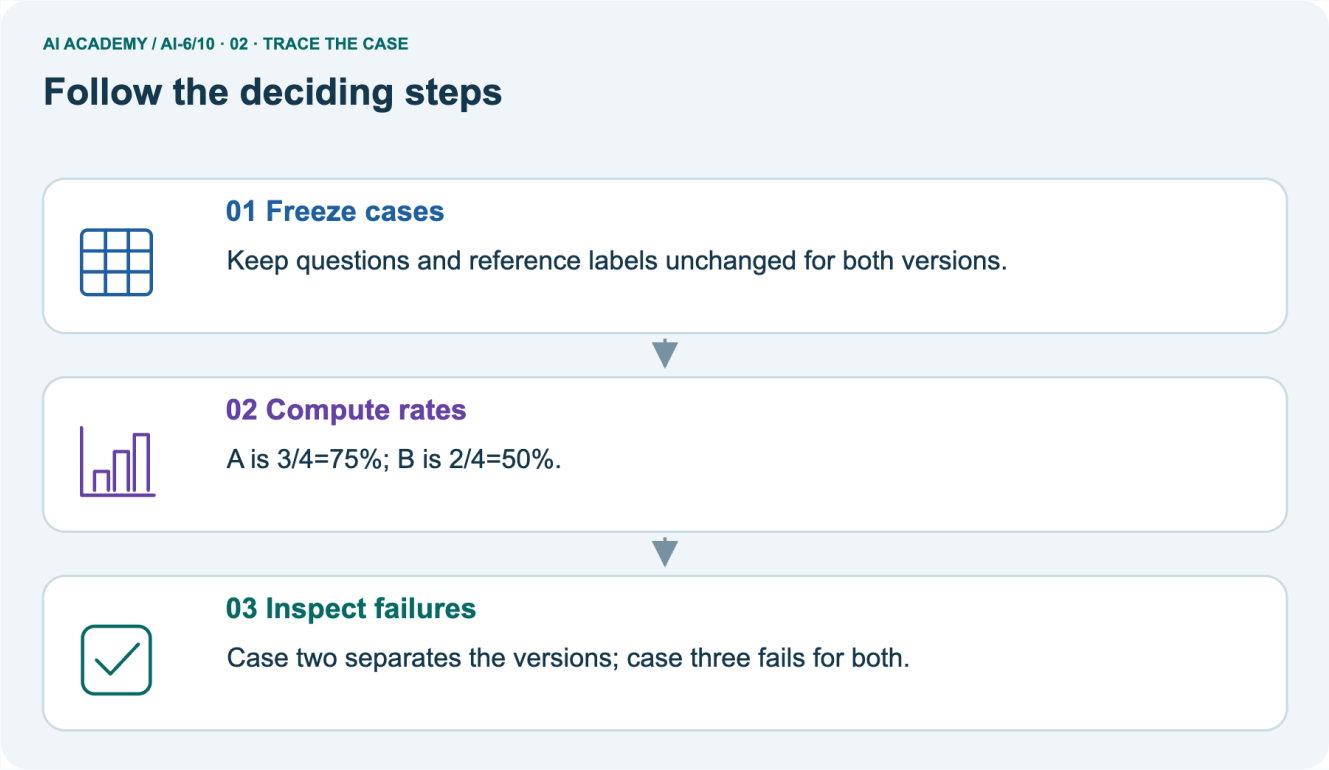
Follow the labelled picture with the matching explanation. Open the vector diagram

Four cases through a scoring pipeline. Two fixed versions answer the same four questions with frozen correct-or-incorrect labels. Supplied facts: A scores [1,1,0,1]; B scores [1,0,0,1].

Each 1 means agreement with the reference answer.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

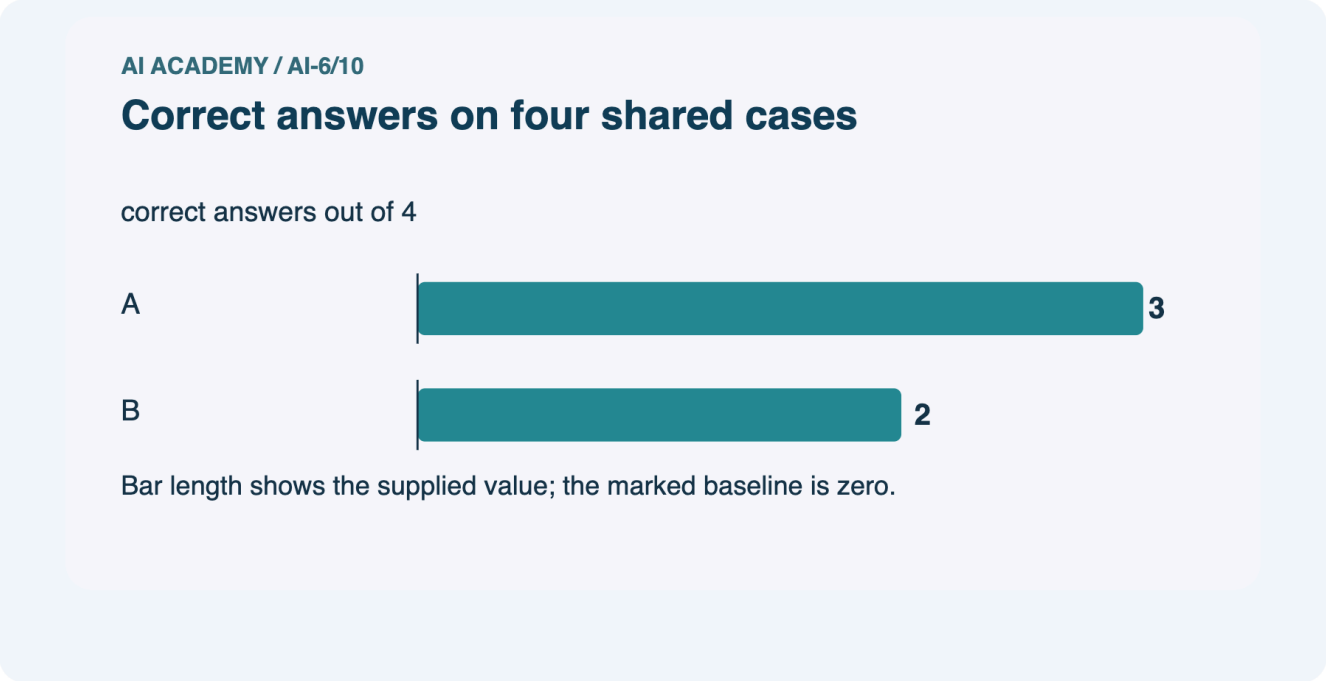


Follow the labelled picture with the matching explanation. Open the vector diagram

Freeze cases: Keep questions and reference labels unchanged for both versions. → Compute rates: A is 3/4=75%; B is 2/4=50%. → Inspect failures: Case two separates the versions; case three fails for both.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Correct answers on four shared cases

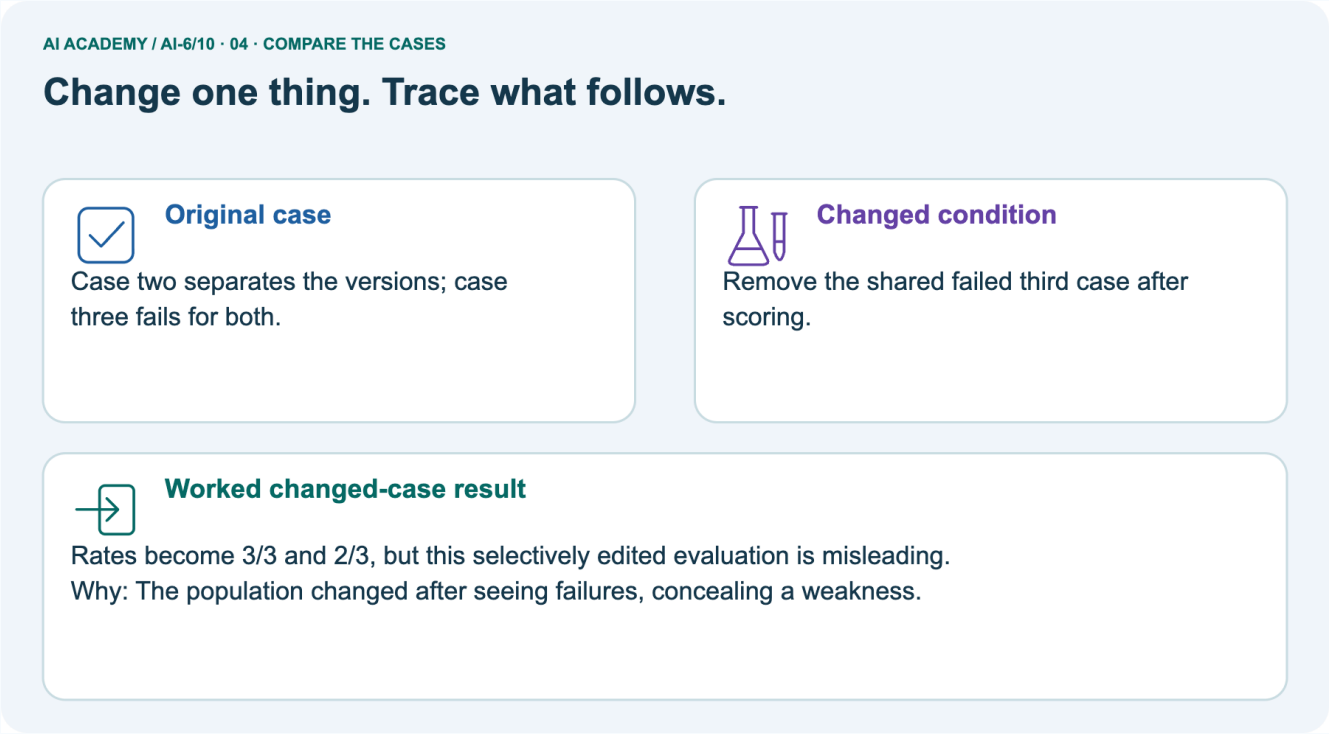


Follow the labelled picture with the matching explanation. Open the vector diagram

Correct answers on four shared cases. Units: correct answers out of 4. A: 3. B: 2.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



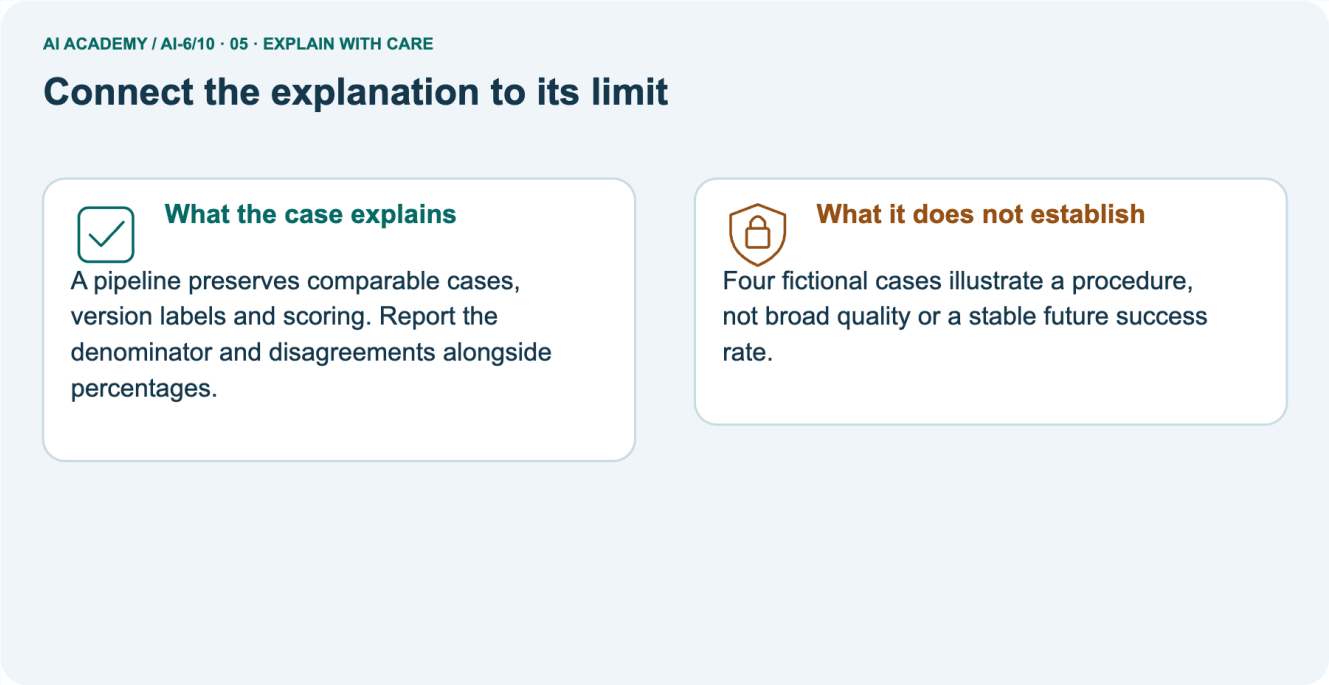
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Case two separates the versions; case three fails for both. Changed condition: Remove the shared failed third case after scoring.

Expected result: Rates become 3/3 and 2/3, but this selectively edited evaluation is misleading. Reason: The population changed after seeing failures, concealing a weakness.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



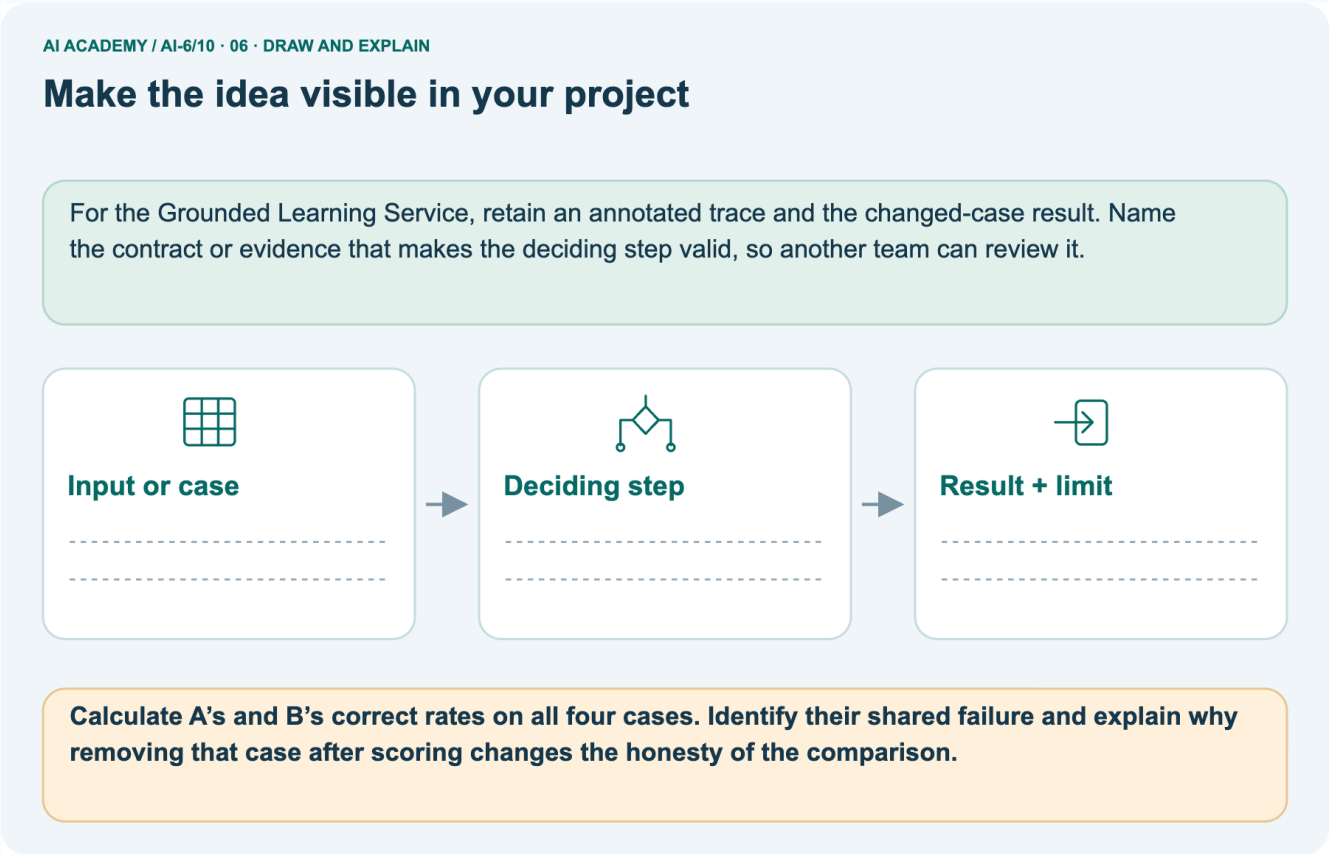
Follow the labelled picture with the matching explanation. Open the vector diagram

A pipeline preserves comparable cases, version labels and scoring. Report the denominator and disagreements alongside percentages.

Limit: Four fictional cases illustrate a procedure, not broad quality or a stable future success rate.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate A's and B's correct rates on all four cases.

Identify their shared failure and explain why removing that case after scoring changes the honesty of the comparison.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.